

HRL International Private Limited

"We Can Do Everything Related To Software Sector Without Any Excuses!"

Entity: Deep-Tech & AVGC Startup
Location: Mangaluru, Karnataka (Beyond Bengaluru Hub)
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Official Product Prototype Dossier & Technical Proof

Government Grant & Incubation Submission (GENESIS EIR / Karnataka ELEVATE / Startup India)

Prototype 1: AI Cinematic Haze — DaVinci Resolve OpenFX Plugin

Hardware-Accelerated Optical Volumetric Atmospheric Scattering Engine

PRODUCTION ALPHA / TESTED

Core Tech Stack: C++17/20, Apple Metal (MSL), NVIDIA CUDA, OpenFX 1.4 API.

AI Depth Model: Monocular Neural Depth Estimation (Depth Anything) via ONNX Runtime.

Color Science: Strict Hollywood ACEScc gamut transforms; 12-bit uncompressed DPX frame rendering.

Performance: Real-time 4K 60FPS local GPU execution with zero cloud latency.

Packaging: Native macOS `.dmg` installer & cross-platform dynamic library binaries.

[0FX Architecture Pipeline Schema]

RAW 12-Bit DPX Sensor Frame → ACEScc Wide Gamut
↓
[ONNX Runtime Neural Depth Estimator (GPU)]
↓ Monocular Z-Depth Map (32-Bit Float Matrix)
[Volumetric Mie Scattering GPU Kernel (CUDA / Metal)]
↓
Real-Time 4K 60FPS Cinema Output Node (0ms Cloud Lag)

Prototype 2: InvigiMatrix — Institutional Examination Optimization Engine

NP-Hard Constraint Satisfaction Solver & Anti-Cheating Dispersion Matrix

LIVE DEPLOYED @ SCEM

Core Algorithm: Combinatorial NP-hard CSP Backtracking with zero-collision guarantees.

Machine Learning: Dynamic K-Means Clustering ($k=3$) + Bayesian student classifier for anti-cheating bench interleaving.

Backend Engine: Native Node.js HTTP server + SQLite 3 with strict ACID transactions.

Deployment: Actively deployed for 100+ faculty duties at Sahyadri College (SCEM).

Exports: High-DPI HTML5 2D Canvas A4 seating arrangement charts.

[Live Institutional Solver Metrics]

- Faculty Constraint Solver: Resolved 120+ duties with 0 collisions.
- Anti-Cheating Matrix: Interleaved 4 branches across 45 halls.
- Transaction Safety: ACID-compliant SQLite 3 WAL Mode.
- UI/UX: Apple Liquid Glass Dynamic Pill Interface.
- Live Repository: github.com/hrlpavan/exam-invigilation-dsa-app

Prototype 3: OmniTransform AI & Voice Studio

Autonomous Multi-Agent Media Orchestration & Neural Voice Synthesis

VALIDATED / LIVE

Voice AI Engine: Sub-75ms streaming neural speech synthesis via ElevenLabs Multilingual v2.

Visualizer: Real-time Chladni resonance acoustic waveform frequency physics.

Agentic Fleet: Autonomous multi-agent orchestration for multi-channel video syndication.

Audience Validation: Powered 2.5M+ organic impressions across creator channels.

IP Protection: Built-in client-side anti-scraping and code obfuscation shields.

[Creator Reach & Production Metrics]

- Total Organic Reach: 2,500,000+ Verified Impressions.
- Ecosystem Hubs: @hrlpremiumstudio, @hrlefx, @hrlflix.
- Streaming Latency: < 75ms Neural Speech Inference.
- Security Accreditation: Deloitte Australia Cyber Defense.

Live Verifiable Repository & Platform Links:

- Live Platform: hrl-brand-seo.vercel.app | Website: hrlpavan.github.io/hrl-international-website-/
- Exam Engine Repo: github.com/hrlpavan/exam-invigilation-dsa-app | Pitch Deck: [HRL_International_Pitch_Deck.pdf](#)